CHINESE ACADEMY of MEDICAL SCIENCES OXFORD INSTITUTE

NUFFIELD DEPARTMENT of MEDICINE

UNIVERSITY OF OXFORD

Basic Immunology Refresher

Online Training Programme 2026

Dannielle Wellington

What does the Immune system do?

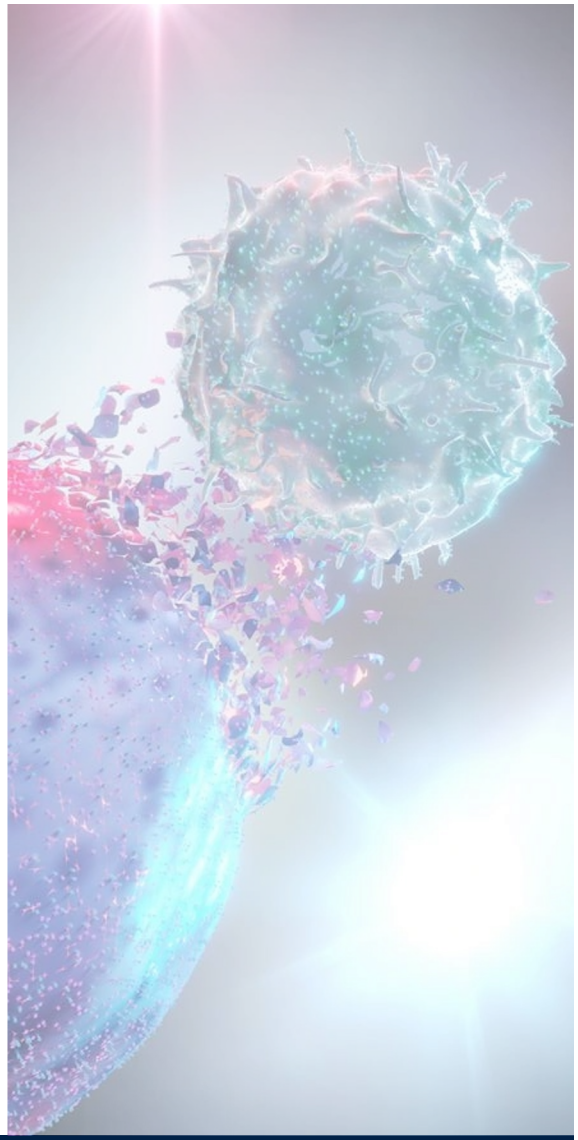
Detect:
recognise infection
or tissue damage

Respond:
Eliminate or contain
the threat

Immune System

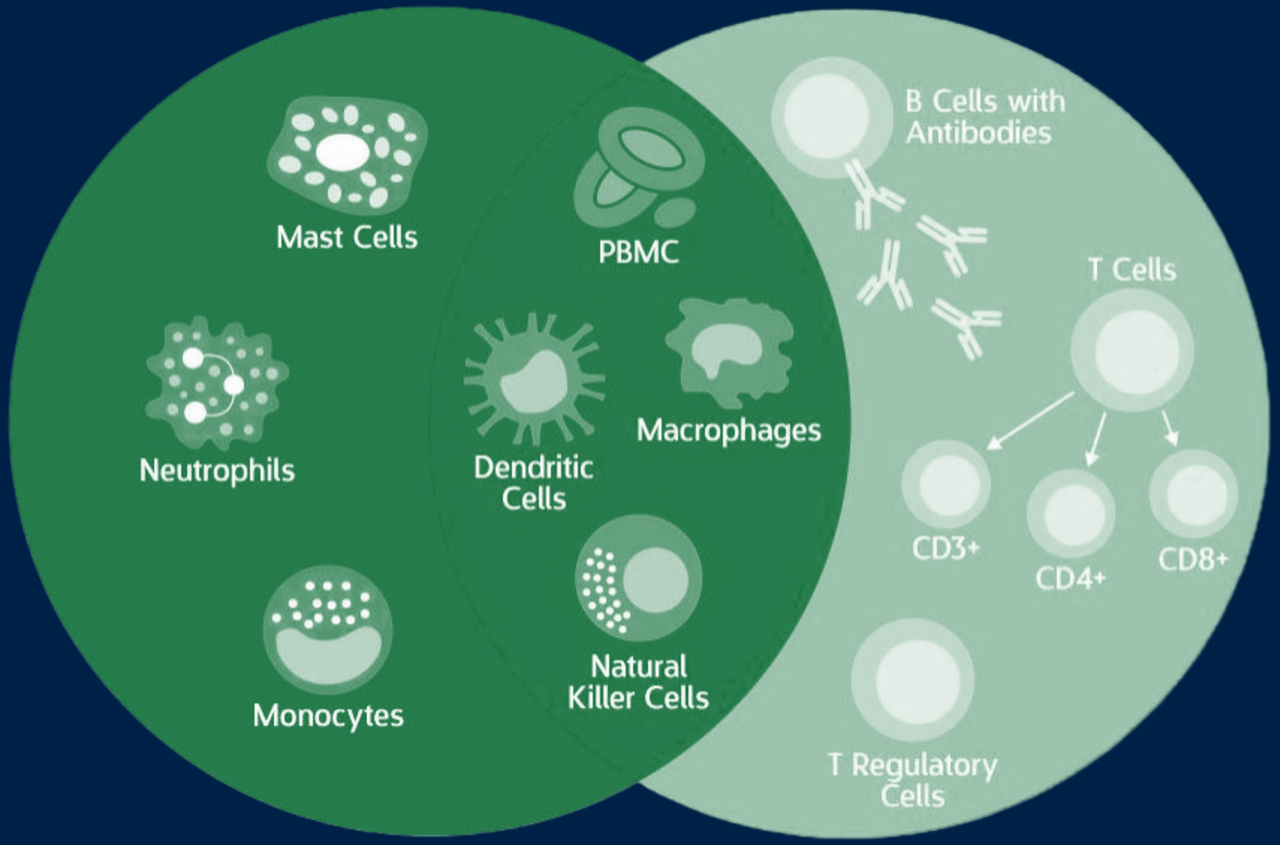
Regulate:
Limit damage to
healthy tissue

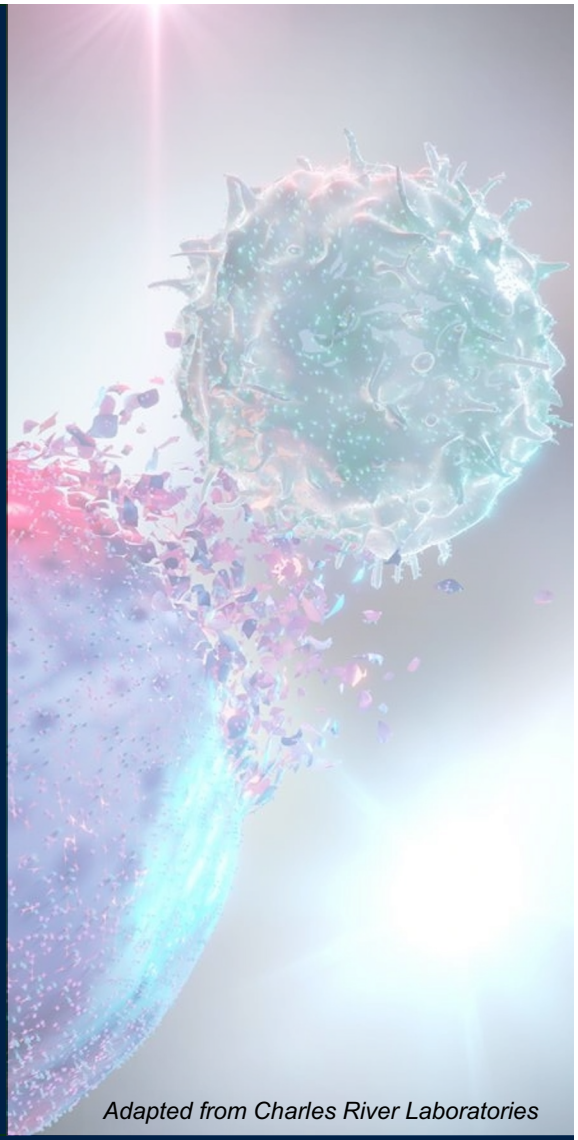
Remember:
Respond more
effectively next time



Immunity is a balance between protection and pathology

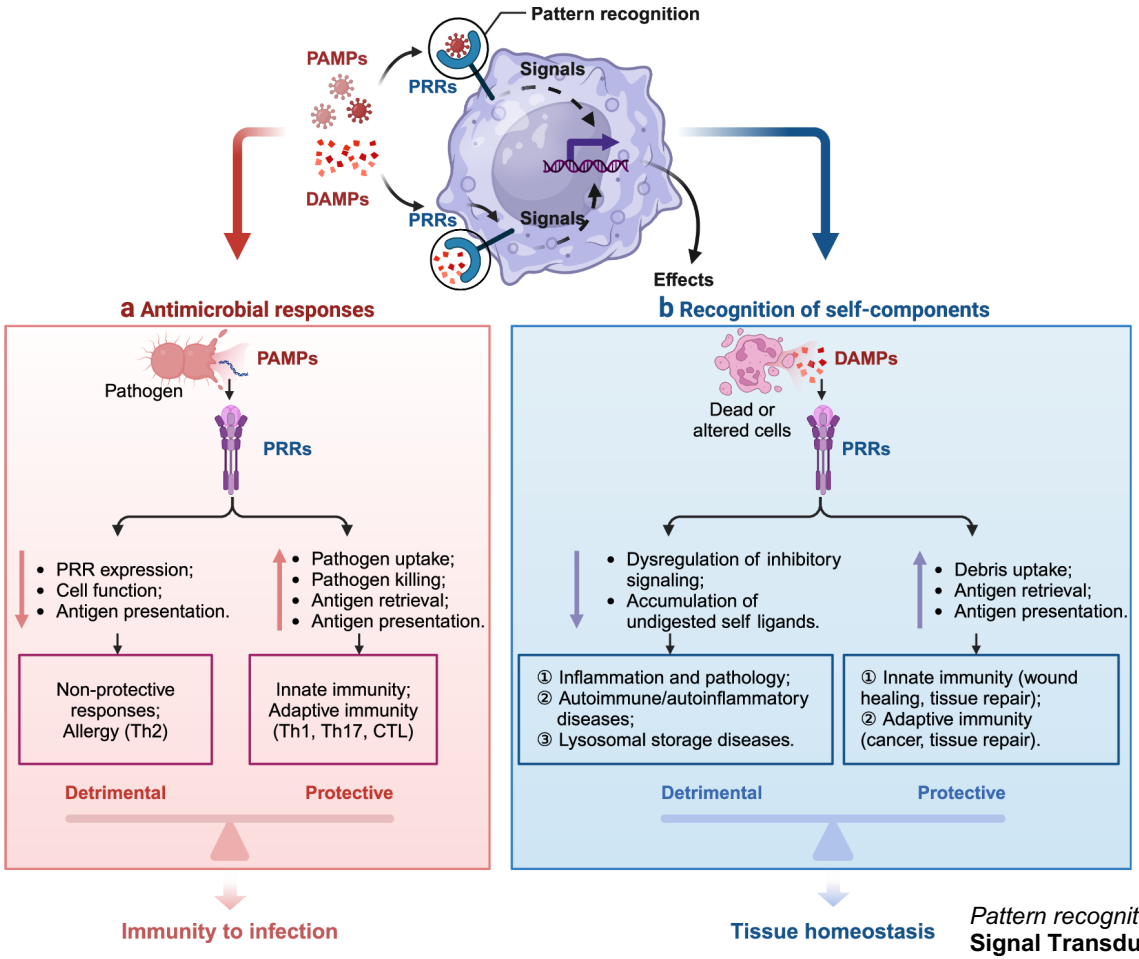
Innate vs Adaptive Immunity

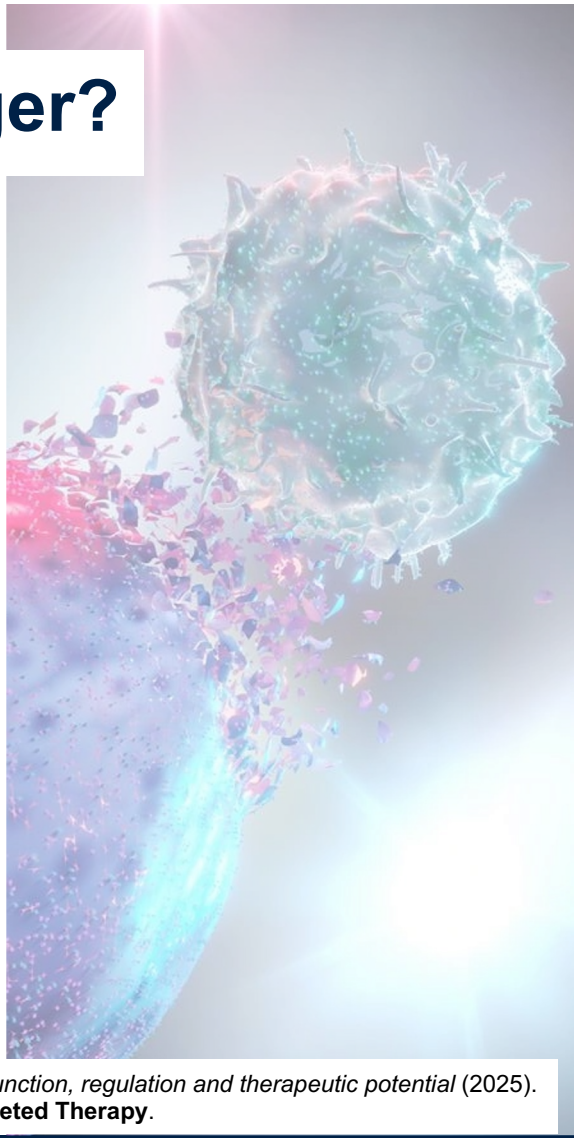




Adapted from Charles River Laboratories

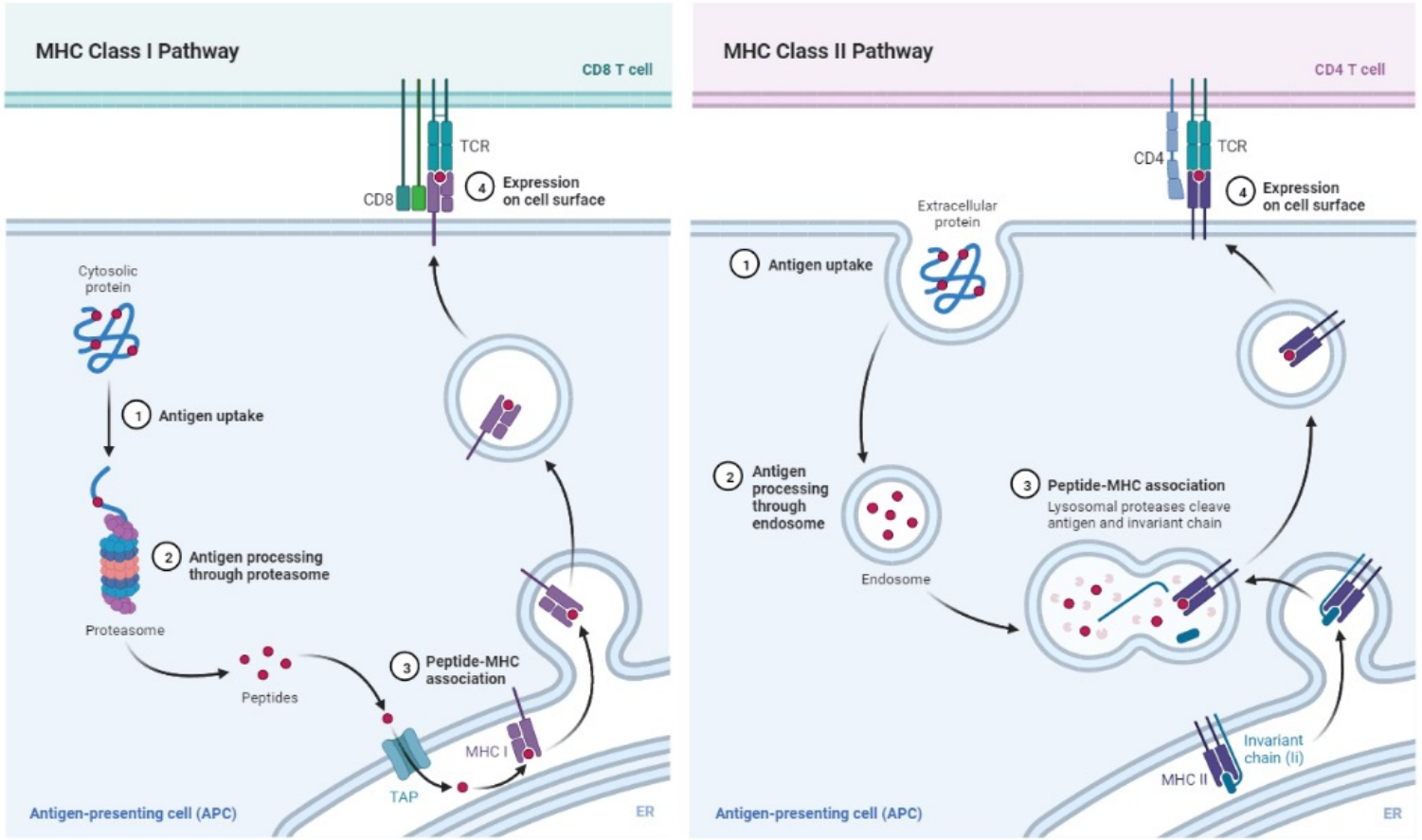
How Does Innate Immunity Recognise Danger?

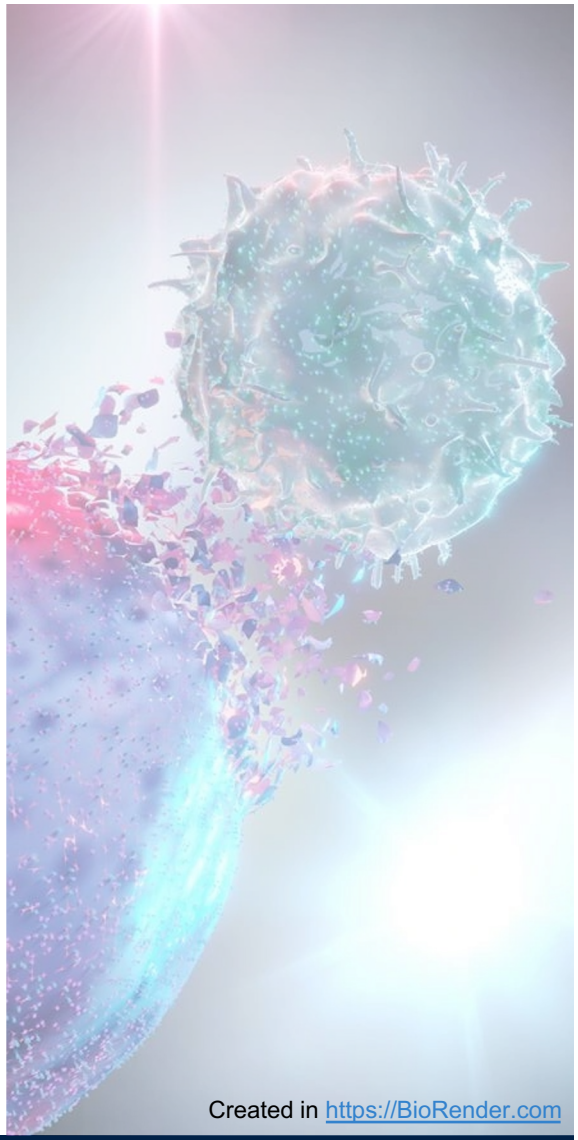




Pattern recognition receptors: function, regulation and therapeutic potential (2025). **Signal Transduction and Targeted Therapy**.

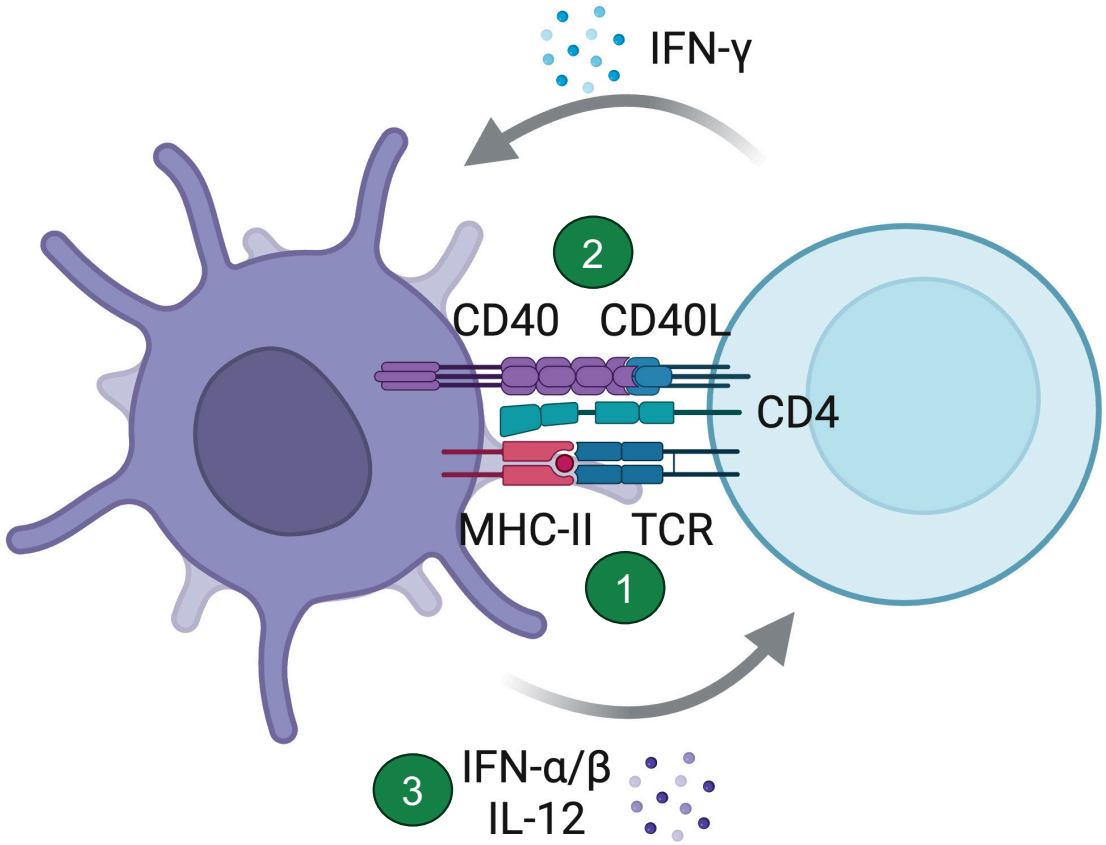
Antigen Presentation: The Critical Bridge

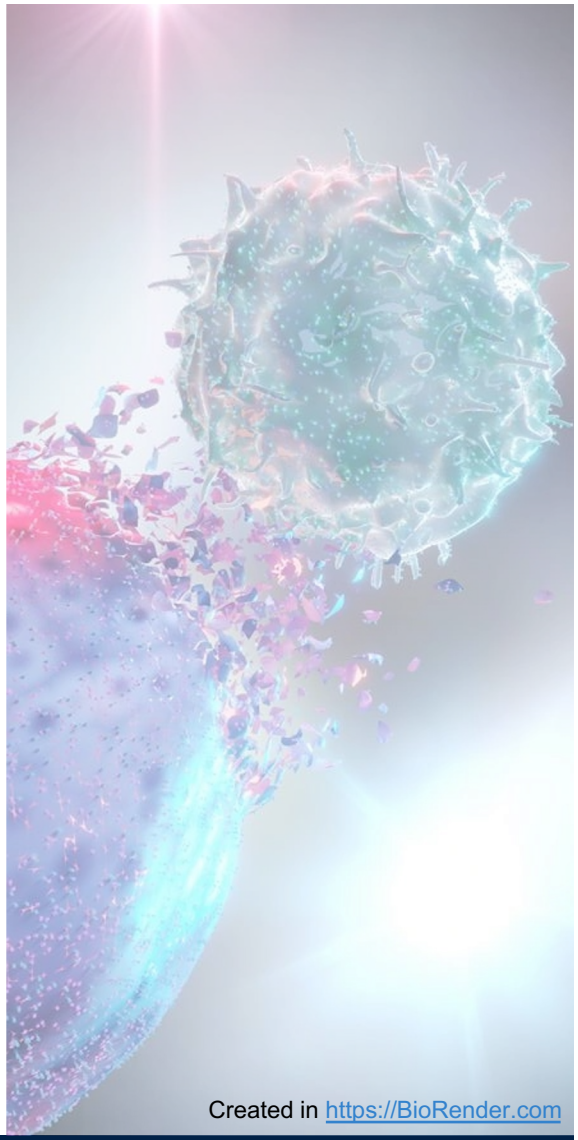




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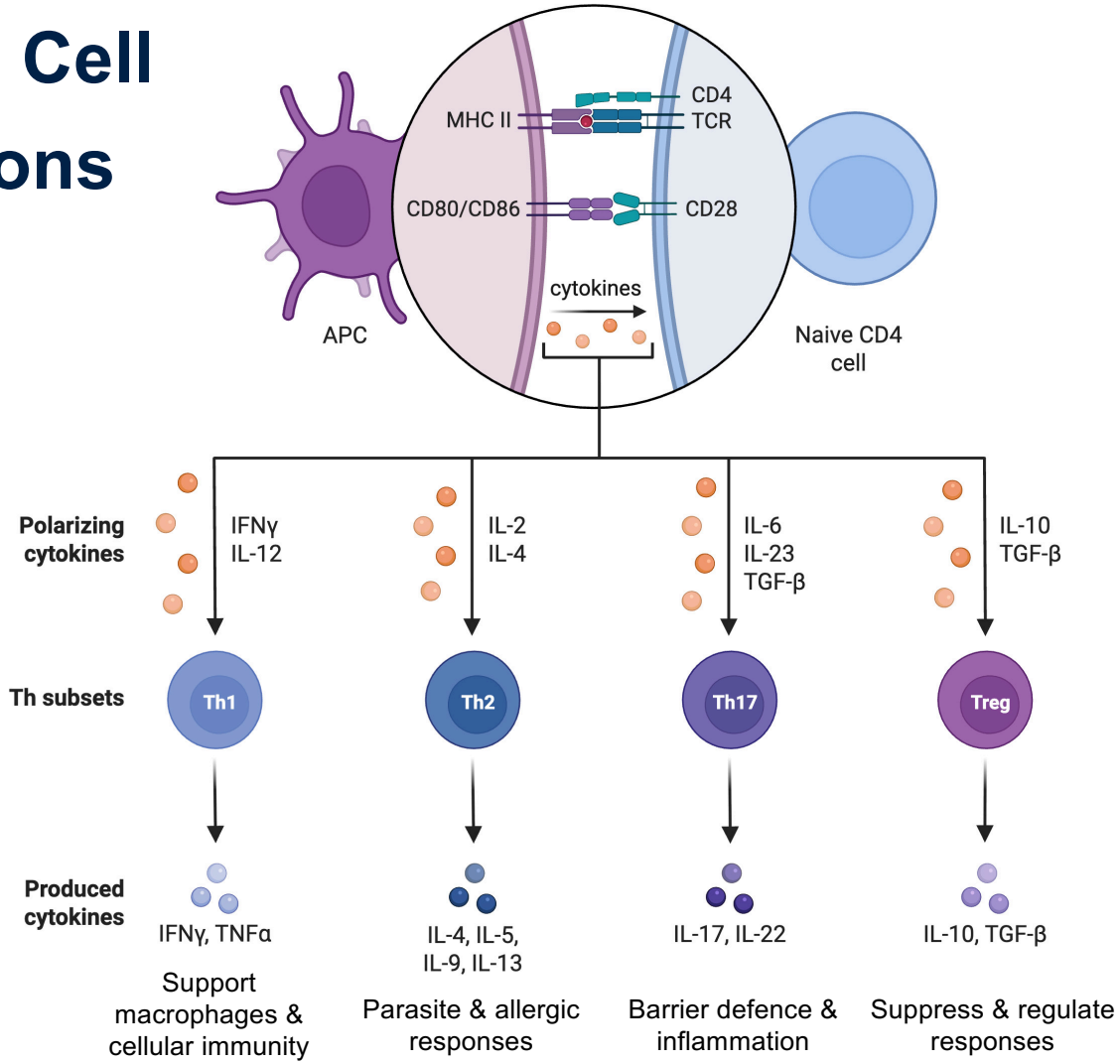
T Cell Activation

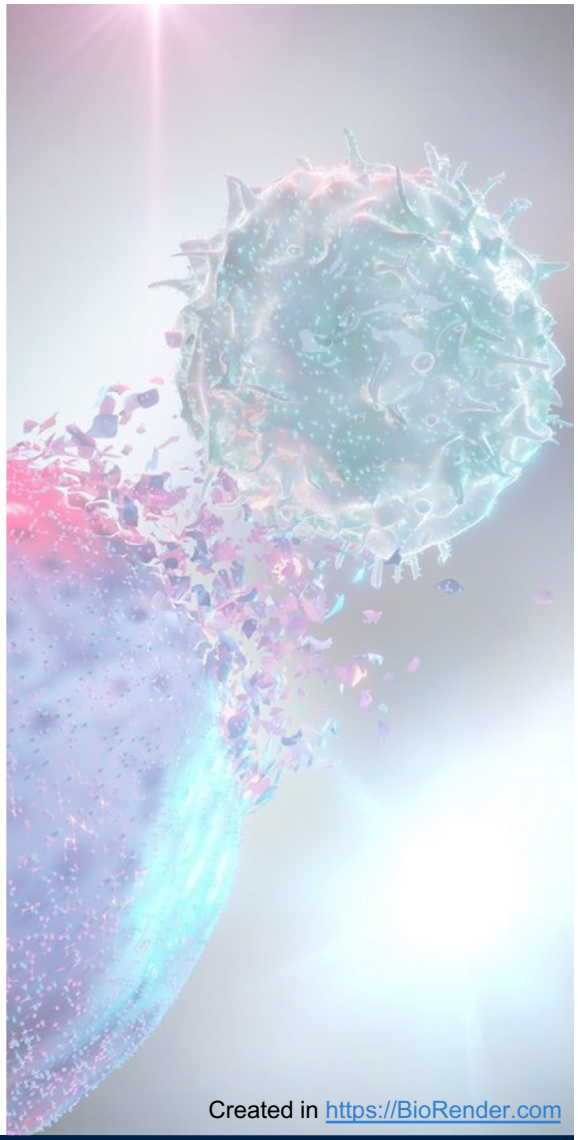




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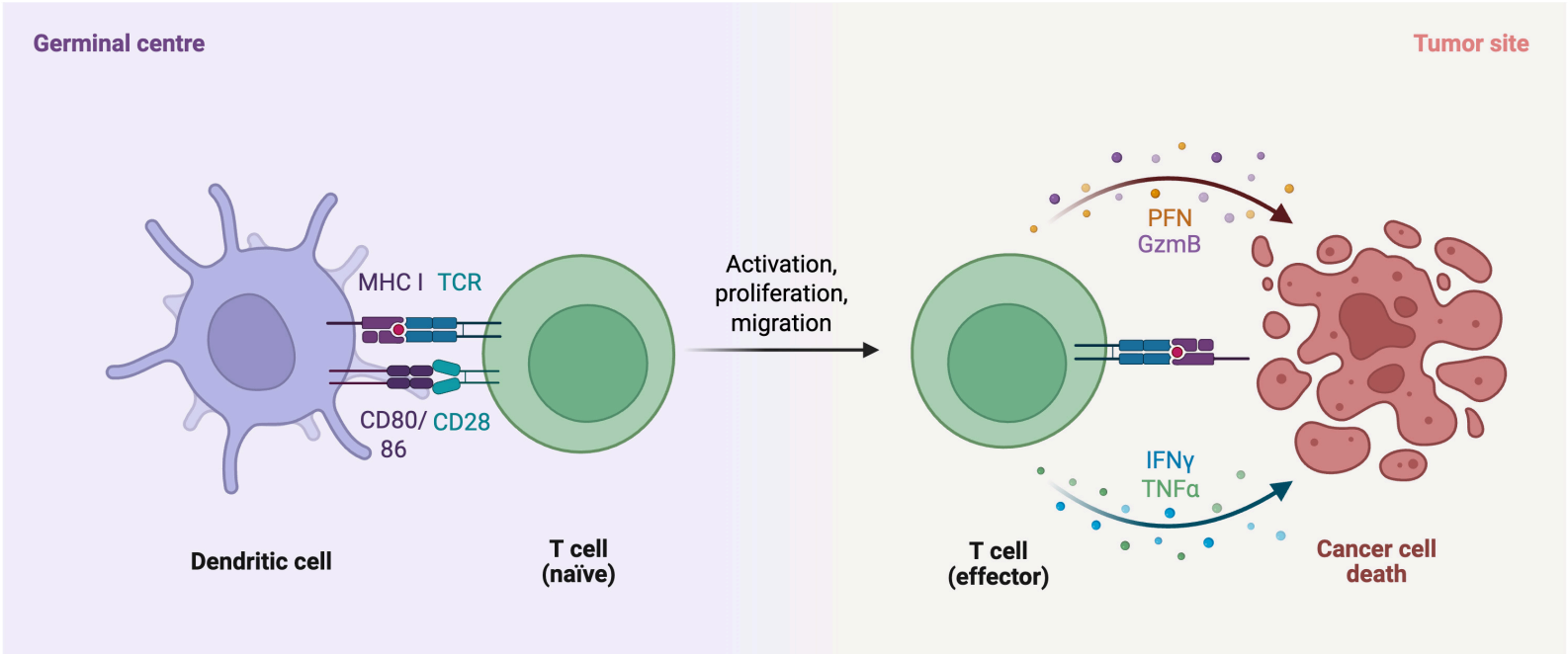
CD4⁺ T Cell Functions

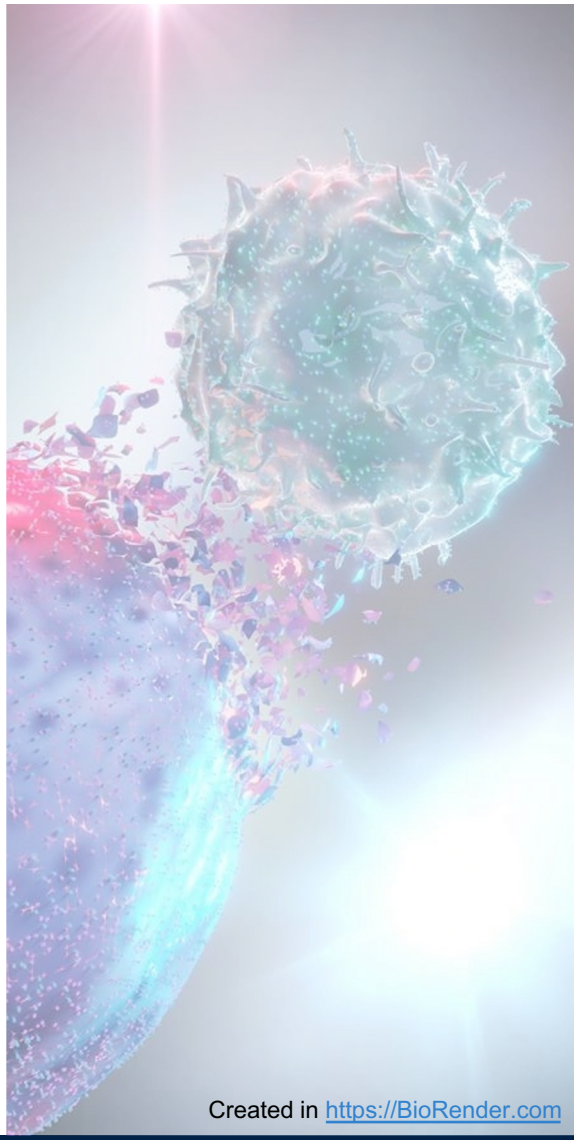




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CD8⁺ T Cell Functions

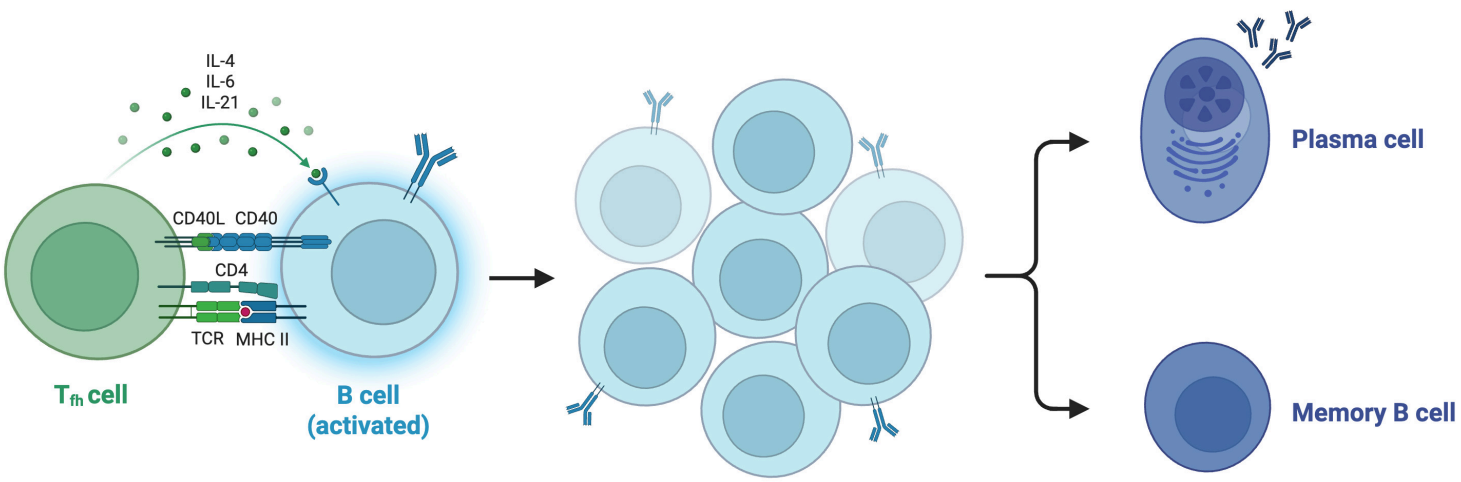


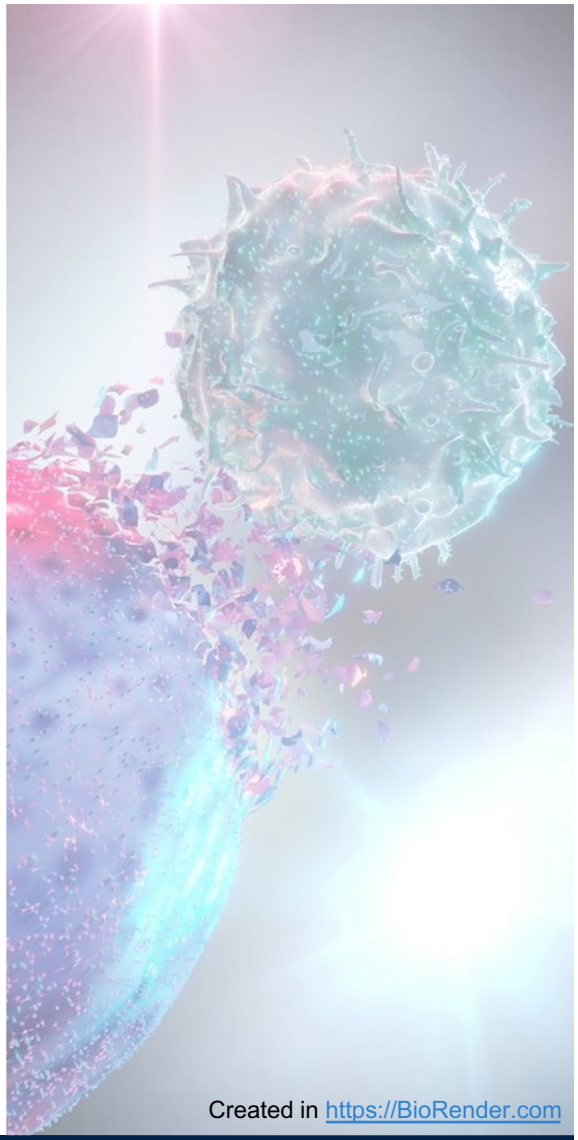


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B Cell Differentiation

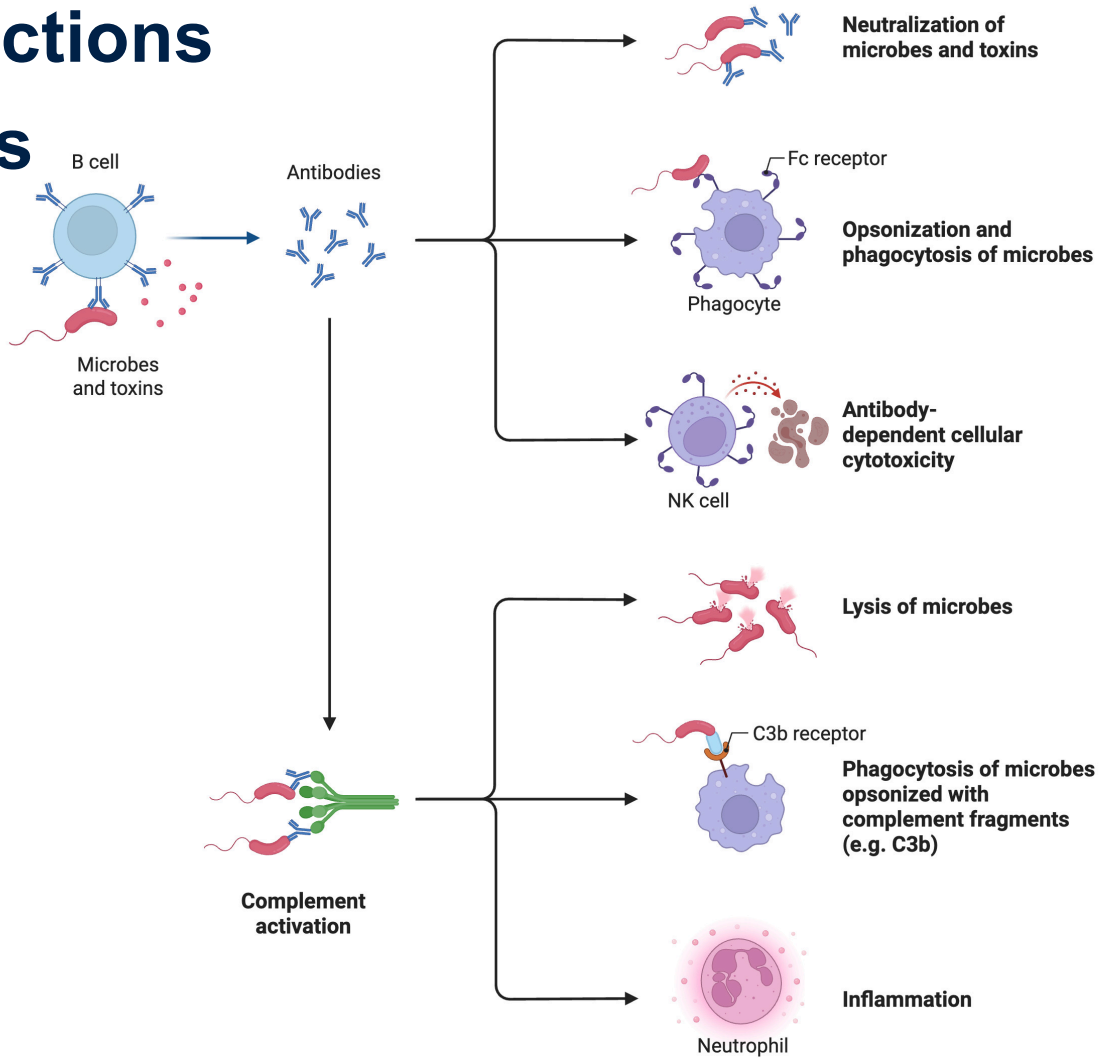
Antigen recognition induces expression of effector molecules by T cell, which activates the B cell → B-cell proliferation → Differentiation to resting memory cells and antibody-secreting plasma cells

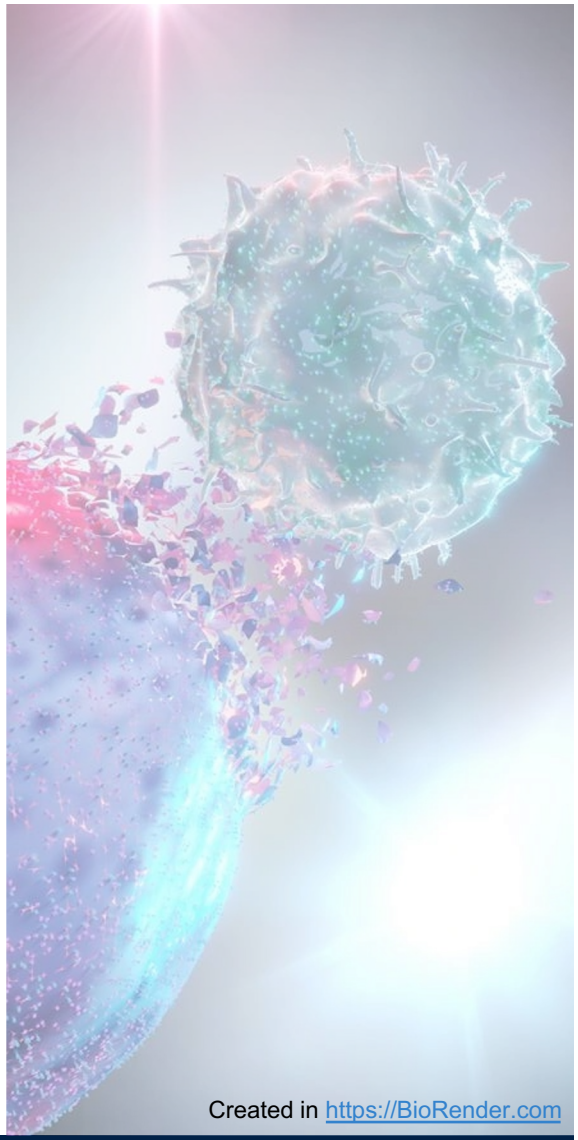




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Effector Functions of Antibodies





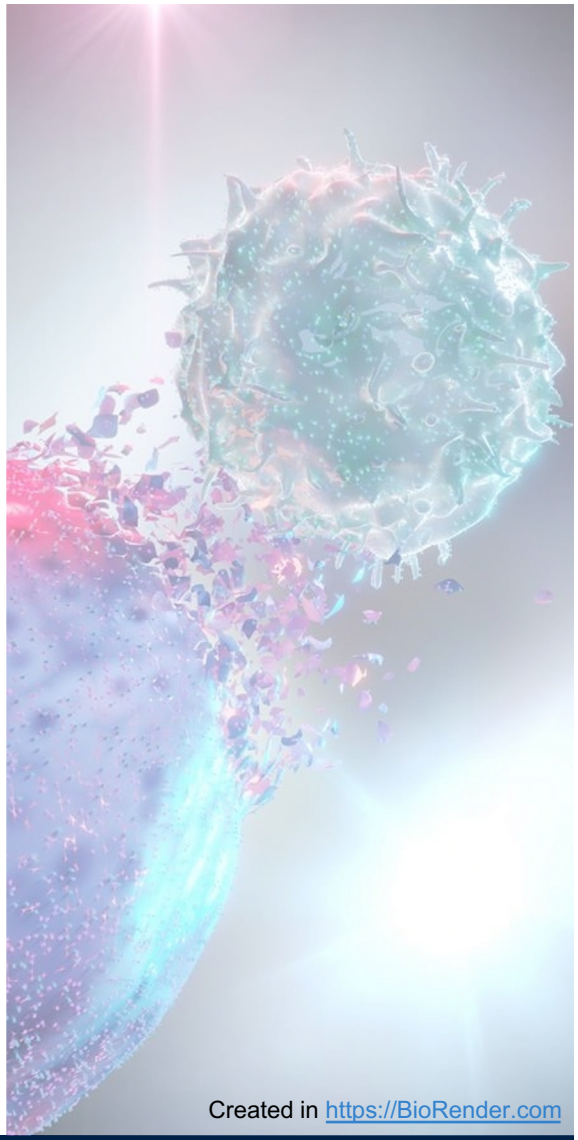
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Table | Antibody Isotypes and their Functions

Classification adapted from [Nature Immunology 2002].

Isotype	Produced by	Function	Site	Clinical
IgM 	Naive B cells/ B1 cells	Complement activation	Blood	Early infection
IgG 	Memory B cells	ADCC, Opsonization	Tissues, Blood	Long-term immunity
IgA 	Plasma cells	Neutralisation	Mucosa	Gut and Lung defense
IgE 	Plasma cells	Mast cell activation	Skin, Mucosa	Allergy, Asthma
IgD 	Naive B cells	Regulation	Upper airways, tonsils	Mainly surface-bound, not secreted

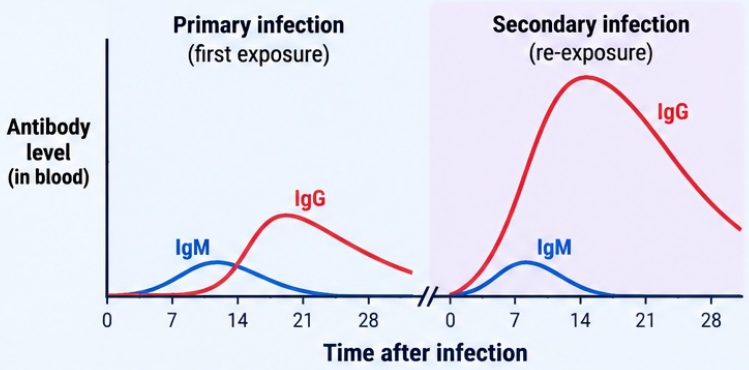
Same antigen specificity + different constant region = different effector function



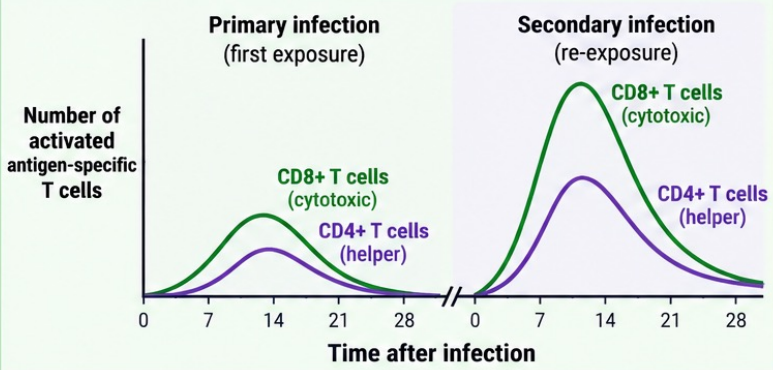
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Secondary immune responses are faster and stronger due to memory B and T cells

A. B cell (antibody) response

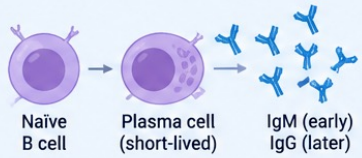


B. T cell response



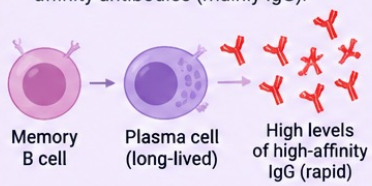
Primary B cell response

- Naïve B cells recognise antigen.
- Clonal expansion and differentiation into plasma cells.
- Initial antibody production (mainly IgM, followed by IgG).



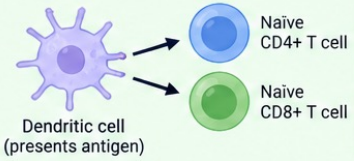
Secondary B cell response

- Memory B cells respond rapidly to the same antigen.
- Faster clonal expansion.
- More plasma cells and higher affinity antibodies (mainly IgG).



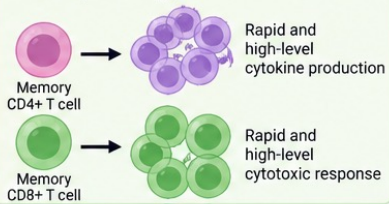
Primary T cell response

- Naïve T cells recognise antigen presented by dendritic cells.
- Clonal expansion and differentiation into effector T cells.
- Slower and lower magnitude response.



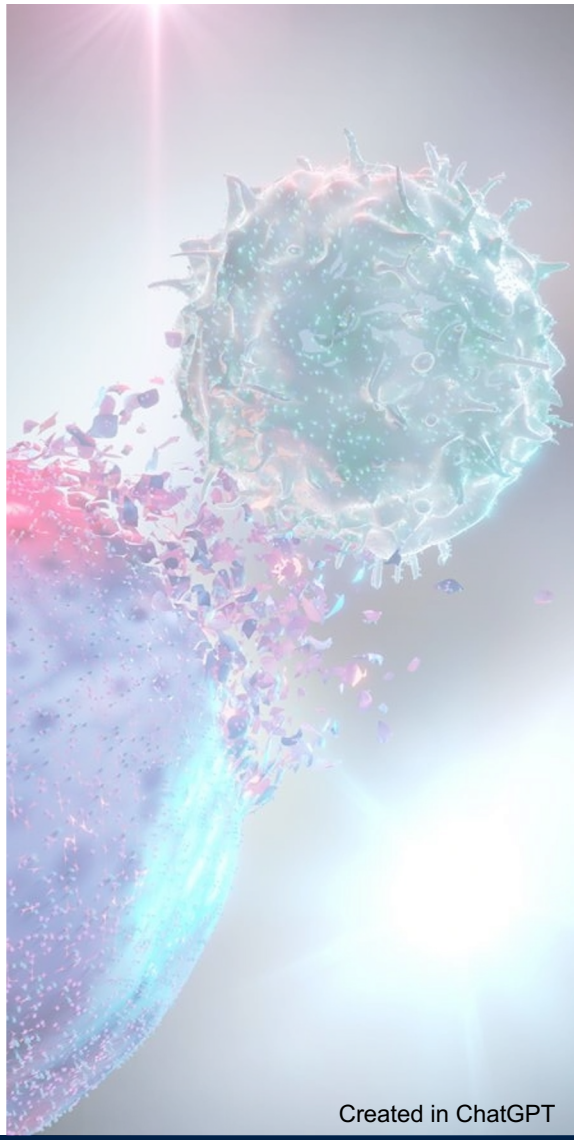
Secondary T cell response

- Memory T cells are rapidly reactivated.
- Faster clonal expansion.
- More effector T cells with enhanced function (e.g. more cytokine production and cytotoxicity).



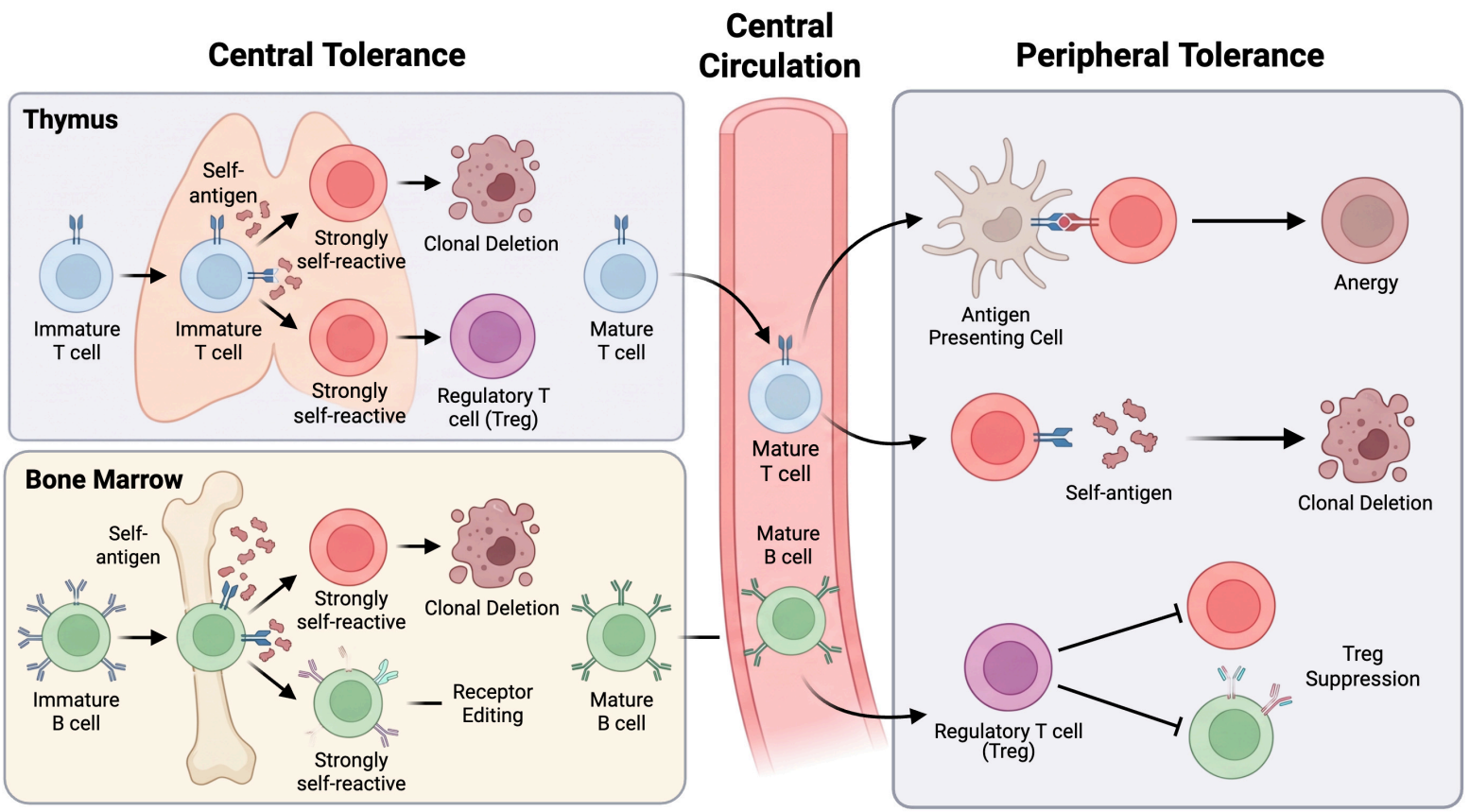
Key reasons secondary responses are faster and stronger

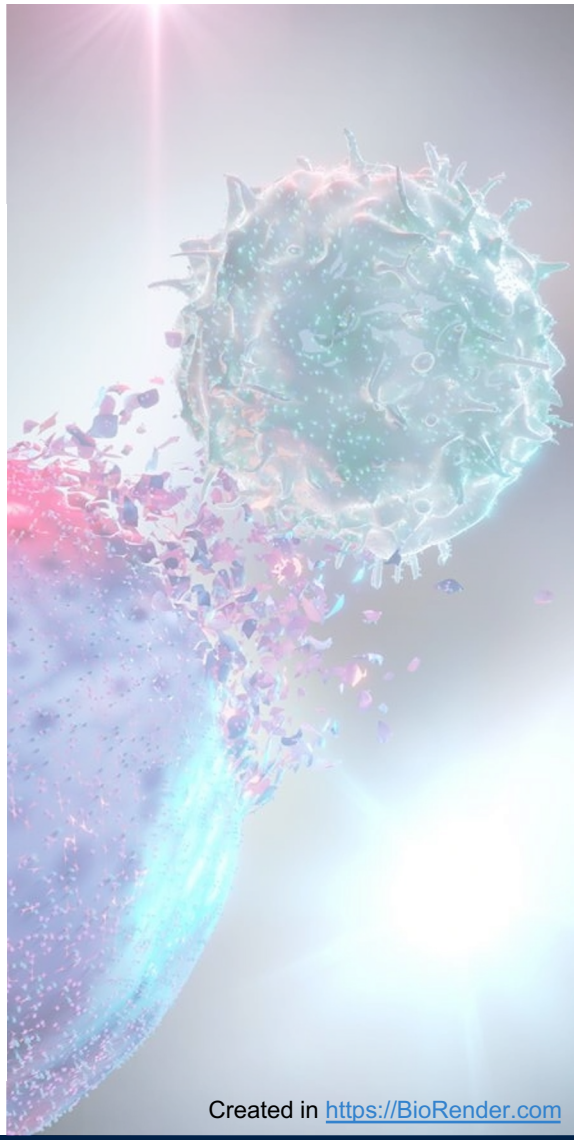
1. **Memory B and T cells**
   Present after the first infection, so do not need to be generated from naïve cells.
2. **Lower activation threshold**
   Memory cells are more easily activated.
3. **Rapid clonal expansion**
   Memory cells proliferate more quickly.
4. **Enhanced function**
   Produce more antibodies (higher affinity IgG) and more effector T cells with greater cytokine production and cytotoxic activity.



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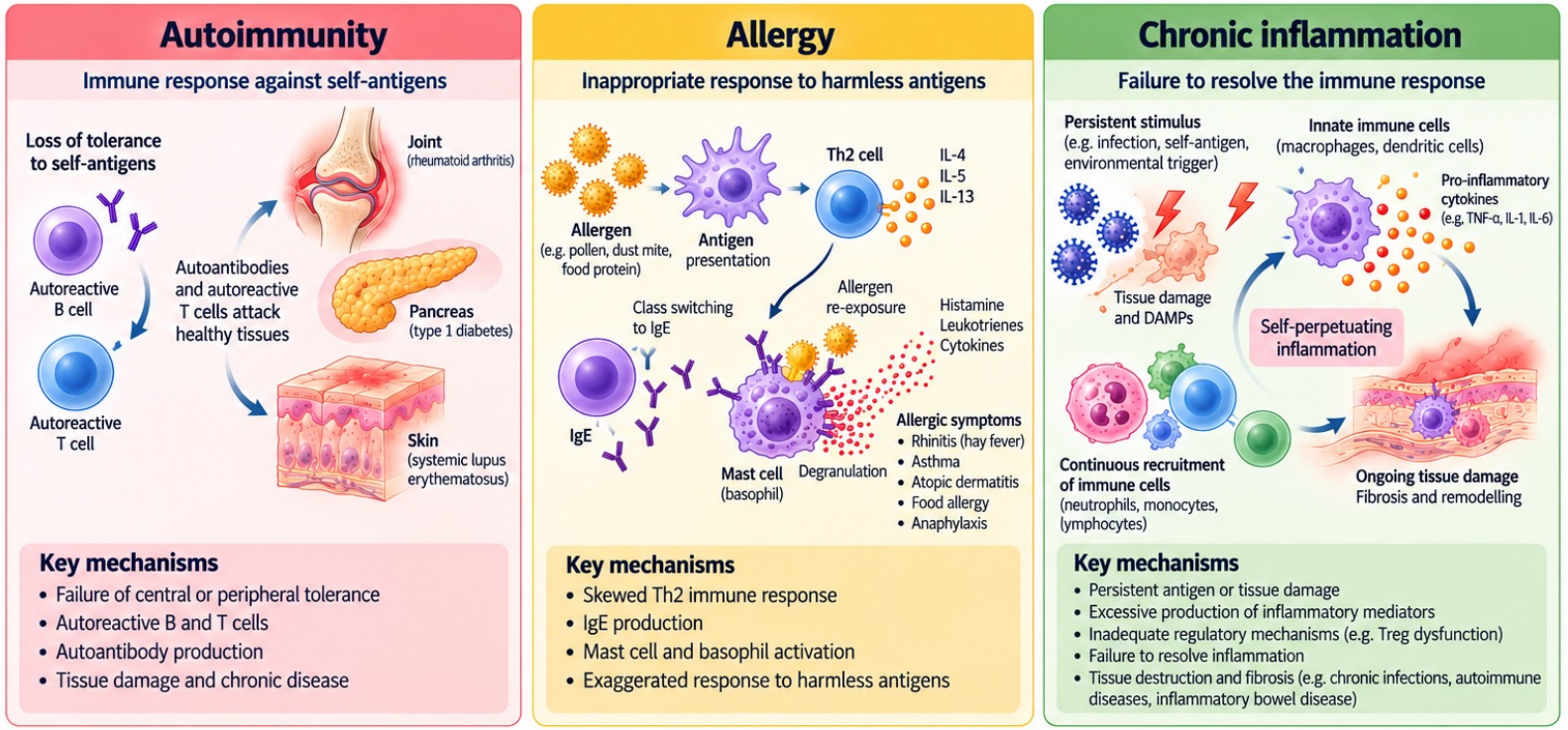
Tolerance Within the System

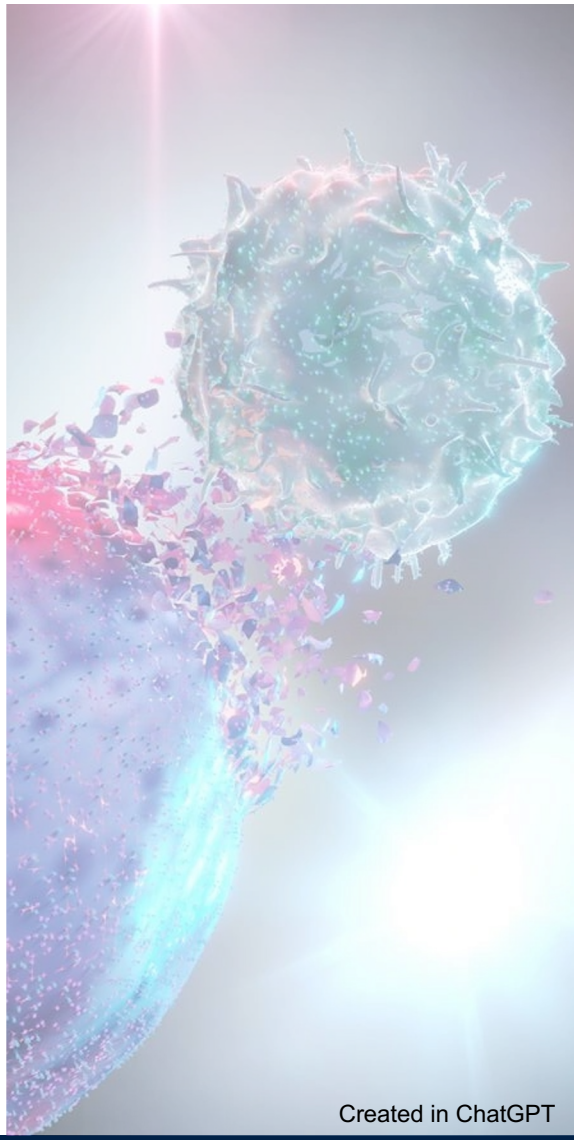




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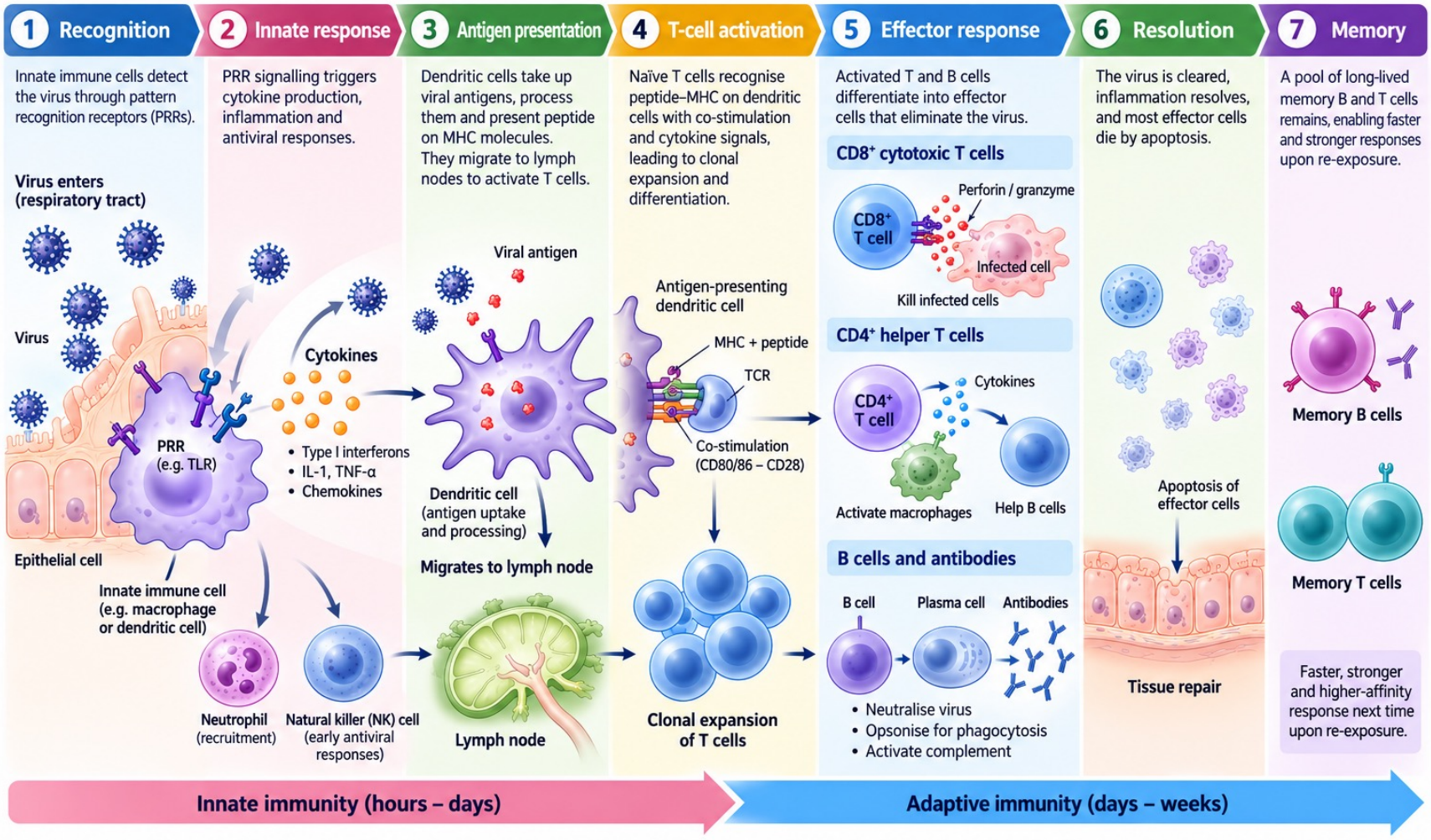
Regulation Failure Can Lead To…





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From Viral Infection to Immune Memory



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Any Questions?